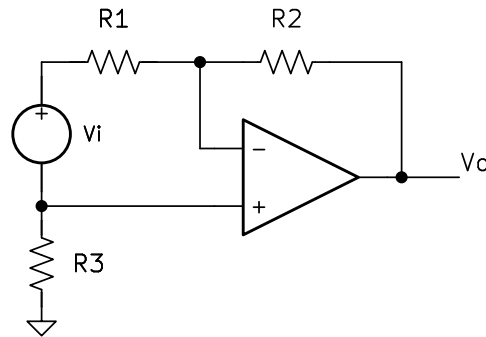


Homework 2

Due February 4th

Warning: In studying the book application of the return ratio analysis and Blackman's formula, you should be aware of errors in book equations 1.70, 1.71, 1.75a, 1.75b, and 1.76. See the book errata on LMS for corrections to these equations.

Problems:



1. Using the circuit as drawn above, which is driven by an *ideal differential voltage source*, do the following steps:
 - (i) Find the feedback factor, b , of the system. Reminder: to do this, you assume ideal conditions.
 - (ii) Solve for the closed loop gain using *exact analysis* while incorporating the effects of r_d , r_o , and a finite a . Do not make assumptions on the relative sizes of these values to R_1 , R_2 , and R_3 . *You do not need to significantly simplify your answer. Once you have $A = V_o/V_i$ in terms using only the circuit parameters, you may stop.*
 - (iii) Solve for the closed loop gain using *return ratio analysis*. *You do not need to significantly simplify your answer. Once you have T and a_{ft} in terms using only the circuit parameters, you may stop.*
 - (iv) Note that the above two steps should produce the exact same equations, although they may not be formatted the same way symbolically. Check your work by simplifying each with $r_d = \infty$ and $r_o = 0 \Omega$. If correct, you should produce the below gain for each.

$$A = -\frac{\frac{R_2+R_3}{R_1}}{1 + \frac{R_1+R_2+R_3}{aR_1}} = -\frac{R_2 + R_3}{R_1 + \frac{R_1+R_2+R_3}{a}}$$

- (v) Using *Blackman's Formula*, find the input impedance, R_i , the output impedance, R_o , and the circuit impedance as seen by R_2 , R_{c2} . For the both R_i and R_{c2} , the “nodes” in the analysis would be each side of the voltage source or resistor, respectively. For ease of analysis, you may assume $r_d \gg R_1, R_2, R_3 \gg r_o$.
2. Book Problem 1.61. Note: P1.15-P1.19 refers to the figures associated with problems 1.15-1.19, not the chapter figures 1.15-1.19
3. ECSE-6050 ONLY: Book Problem 1.64.
4. Book Problem 1.74.
5. Book Problem 1.77. Note: Fig. P1.17, NOT Fig. 1.17. Instead of using the input $\pm 9V$ sine wave as requested by the book, use a $\pm 25V$ triangular wave.